2 Fig. 2.1 shows a block of mass M_1 = 4.0 kg released from a vertical height of 6.0 m on a curved frictionless track. It slides down the track and makes a head-on elastic collision with a block of mass M_2 = 9.0 kg that is initially at rest.

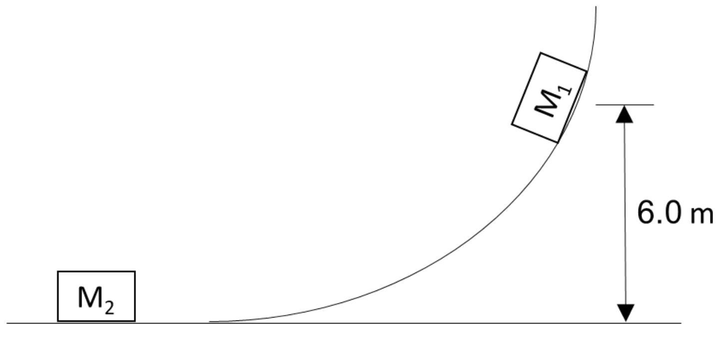

Fig. 2.1

(a) State the principle of conservation of linear momentum.

The <u>total momentum of a system remains constant</u> provided <u>no external resultant force</u> acts on the system.

……………………………………………………………………………………………… [1]

(b) Calculate the velocity of M_1 just after it collision with M_2.

loss in GPE = gain in KE

$M_1gh = \frac{1}{2} M_1u_1^2$ [M1]

$u_1 = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 6.0}$

$= 10.8$ m s^{-1}

By Conservation of Momentum

$M_1u_1 + M_2u_2 = M_1v_1 + M_2v_2$

$(4.0)(10.8) = (4.0)v_1 + (9.0)v_2$

Since is elastic collision, using relative speed of approach/separation

$10.8 - 0 = -(v_1 - v_2)$

$10.8 + v_1 = v_2$

$(4.0)(10.8) = (4.0)v_1 + (9.0)(10.8 + v_1)$

Therefore $v_1 = -4.15$ (-4.17) m s^{-1}

velocity of M_1 just after collision = ……………………. m s^{-1} [4]

(c) Calculate the maximum height to which M_1 rises after the collision.

loss in KE = gain in GPE

$\frac{1}{2} M_1v_1^2 = M_1gh$

$h = \frac{1}{2} v_1^2/g = \frac{1}{2} (4.15)^2/(9.81)$

$= 0.878$ (0.886) m

maximum height M_1 rises after collision = ……………………. m [2]

	(d)	Sketch a graph, on the given axes in Fig. 2.2, to show how the velocity of M_1 varies from the time of its release to the time it reaches maximum height on its return.	[3]

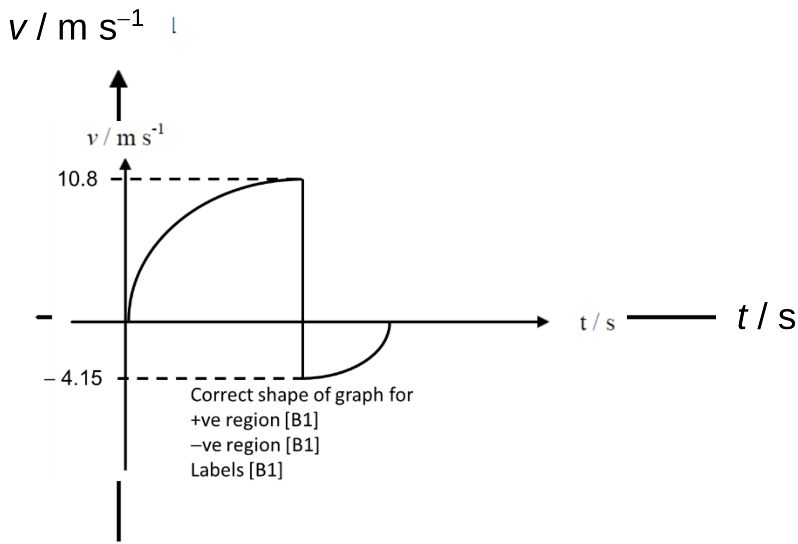

Fig. 2.2

3	**(a)**	**(i)**	Explain what is meant by the term *escape speed*.